Install ARCH Linux with encrypted file-system and UEFI

The official installation guide (https://wiki.archlinux.org/index.php/Installation_Guide) contains a more verbose description.

Download the archiso image from https://www.archlinux.org/

Copy to a usb-drive

dd if=archlinux.img of=/dev/sdX bs=16M && sync # on linux

Boot from the usb. If the usb fails to boot, make sure that secure boot is disabled in the BIOS configuration.

Set swedish keymap ('-' is on '+'-key)

loadkeys dk-latin1

This assumes a wifi only system...

wifi-menu

Create partitions

cgdisk /dev/sdX 1 100MB EFI partition # Hex code ef00 2 250MB Boot partition # Hex code 8300 3 100% size partiton # (to be encrypted) Hex code 8300

mkfs.vfat -F32 /dev/sdX1 mkfs.ext2 /dev/sdX2

Setup the encryption of the system

cryptsetup -c aes-xts-plain64 -y --use-random luksFormat /dev/sdX3 cryptsetup luksOpen /dev/sdX3 luks

Create encrypted partitions

This creates one partions for root, modify if /home or other partitions should be on separate partitions

pvcreate /dev/mapper/luks vgcreate vg0 /dev/mapper/luks lvcreate --size 8G vg0 --name swap lvcreate -l +100%FREE vg0 --name root

Create filesystems on encrypted partitions

mkfs.ext4 /dev/mapper/vg0-root mkswap /dev/mapper/vg0-swap

Mount the new system

mount /dev/mapper/vg0-root /mnt # /mnt is the installed system swapon /dev/mapper/vg0-swap # Not needed but a good thing to test mkdir /mnt/boot mount /dev/sdX2 /mnt/boot mkdir /mnt/boot/efi mount /dev/sdX1 /mnt/boot/efi

Adjust mirrors

nano /etc/pacman.d/mirrorlist

Install the system also includes stuff needed for starting wifi when first booting into the newly installed system

Unless vim and zsh are desired these can be removed from the command

pacstrap /mnt base base-devel grub-efi-x86_64 fish git efibootmgr dialog wpa_supplicant

'install' fstab

genfstab -pU /mnt >> /mnt/etc/fstab

Make /tmp a ramdisk (add the following line to /mnt/etc/fstab)

tmpfs /tmp tmpfs defaults,noatime,mode=1777 0 0

Change relatime on all non-boot partitions to noatime (reduces wear if using an SSD)

Enter the new system

arch-chroot /mnt /bin/bash

Setup system clock

ln -sf /usr/share/zoneinfo/Europe/Copenhagen /etc/localtime hwclock --systohc --utc

Set the hostname

echo MYHOSTNAME > /etc/hostname

Update locale

echo LANG=en_US.UTF-8 >> /etc/locale.conf echo LANGUAGE=en_US >> /etc/locale.conf echo LC_ALL=C >> /etc/locale.conf

Set password for root

passwd

Add real user remove -s flag if you don't whish to use zsh

useradd -m -g users -G wheel -s /usr/bin/fish MYUSERNAME

passwd MYUSERNAME

Configure mkinitcpio with modules needed for the initrd image

vim /etc/mkinitcpio.conf

Add 'ext4' to MODULES

Add 'encrypt' and 'lvm2' to HOOKS before filesystems

Regenerate initrd image

mkinitcpio -p linux

Setup grub

grub-install In /etc/default/grub edit the line GRUB_CMDLINE_LINUX to GRUB_CMDLINE_LINUX="cryptdevice=/dev/sdX3:luks:allow-discards" then run: grub-mkconfig -o /boot/grub/grub.cfg

Enable Network Manager as service

systemctl enable NetworkManager.service

Make keyboard layout persistent

localectl set-x11-keymap dk

Enable sudo for user

visudo

Uncomment:

%wheel ALL=(ALL) ALL

Install Yay, first change to user, then:

sudo pacman -Syu git clone https://aur.archlinux.org/yay.git cd yay makepkg -si

Install yadm

yay -S yadm-git

Pull settings from git

yadm clone git@github.com:morten-b/dotfiles.git

Install apps from lists

cat pacman.lst | xargs pacman -S --needed --noconfirm cat aur.lst | xargs yay -S --needed --noconfirm

Exit new system and go into the cd shell

exit

Unmount all partitions

umount -R /mnt swapoff -a

Reboot into the new system, don't forget to remove the cd/usb

reboot